

Study Notes - Limits

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Module 1: The “Why” of Calculus and the Concept of Limits

Part 1: The Big Picture – Why does Calculus exist?

Math isn't just about memorizing rules; it's about giving us tools to explain the universe.

- **Algebra is for a STATIC world.** It works perfectly when things stay exactly the same. If you drive 60 miles per hour for 2 hours, algebra tells you that you traveled 120 miles. It calculates *averages* over a period of time.
- **Calculus is for a DYNAMIC world.** The real world is constantly changing. You hit the brakes, you speed up, rockets burn fuel, stocks go up and down. Calculus allows us to freeze time and calculate the exact rate of change at a single, infinitesimal millisecond.

Key Takeaway: Algebra gives you the *average* speed of your whole road trip. Calculus is the math that makes your car's *speedometer* work right now.

Part 2: What is a Limit?

Before we can calculate change at a frozen millisecond, we need a new mathematical tool: **The Limit**.

A limit simply asks: “**As we get really, really close to a certain input, what is the output getting close to?**”

Imagine walking toward the edge of a cliff. You can't actually step *off* the edge (in math, dividing by zero is stepping off the cliff—it's “undefined”). However, you can get infinitesimally close to the edge. The limit is simply the exact location of that edge.

In calculus, we write a limit like this:

$$\lim_{x \rightarrow a} f(x)$$

How to read it: “The limit as x approaches a , of the function.”

Part 3: The 3 Rules for Solving Limits

Rule #1: Direct Substitution (Just plug it in!) Most of the time, to find out where a function is headed, you just replace the x with your number.

- *Problem:*

$$\lim_{x \rightarrow 2} (x + 3)$$

- *Step:* Substitute 2 for x .
- *Answer:* $2 + 3 = 5$

Rule #2: The Constant Rule If there is no x to plug your number into, the limit is just the number itself. Think of driving on a perfectly flat road; no matter how far forward you move, your elevation doesn't change.

- *Problem:*

$$\lim_{x \rightarrow 10} 42$$

- *Answer:* 42

Rule #3: The “Zero in the Denominator” Trap (Factor & Cancel) Sometimes, plugging the number in breaks the math. You are never allowed to divide by zero. If you plug a number in and get $\frac{0}{0}$, it means there is a “hole” in the graph. We have to use algebra to fix it.

- *Problem:*

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

- *The Trap:* If we plug in 3, we get $\frac{3^2 - 9}{3 - 3} = \frac{0}{0}$. Math breaks.
- *The Fix (Difference of Squares):* Change $x^2 - 9$ into $(x - 3)(x + 3)$.
- *Rewrite:*

$$\lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{x - 3}$$

- *Cancel:* Cross out the $(x - 3)$ on the top and bottom. You are left with just $x + 3$.
 - *Solve:* Now plug in the 3. $3 + 3 = 6$.
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Part 4: Real-World Application – The Drone Speed Trap

The Scenario: You are flying a drone straight forward. The distance it travels in feet is modeled by the equation **Distance** = x^2 (where x is seconds). You want to find its *exact* speed at exactly the 3-second mark.

Why Algebra Fails: Speed is $\frac{\text{Distance}}{\text{Time}}$. To find the speed at exactly 3 seconds, we have to look at a time window of zero seconds.

- Time elapsed: $3 - 3 = 0$ seconds.
- Distance moved in that frozen moment: $9 - 9 = 0$ feet.
- Speed = $\frac{0}{0}$. Algebra says the speed is undefined.

How Calculus Saves the Day: Instead of freezing time, we use a limit to ask: “*What is the speed as the time interval gets really, really close to 3?*”

- Distance at time x : x^2
- Distance at 3 seconds: 9
- Time interval: $x - 3$

We set up our limit:

$$\text{Speed} = \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

Using **Rule #3** from above, we factor the top, cancel out the $(x - 3)$, and are left with $\lim_{x \rightarrow 3} (x + 3)$. We plug in 3, and get **6**.

Conclusion: At exactly the 3-second mark, the drone was traveling at exactly 6 feet per second.